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ercot
Your Power. Our Promise.

TAYLOR
2705 West Lake Dr.
Taylor, Texas 76574
T: 512-248-3000
F: 512-225-7079

AUSTIN
8000 Metropolis Dr.
Bldg. E, Suite 100
Austin, Texas 78744
T: 512-225-7000
F: 512-225-7079

ercot.com

October 20, 2025

Public Utility Commission of Texas
Chairman Thomas J. Gleeson
Commissioner Kathleen Jackson
Commissioner Courtney K. Hjaltman
1701 N. Congress Avenue
Austin, TX 78711

Re: PUC Project No. 55999, *Reports of the Electric Reliability Council of Texas* – Presentation Regarding the ERCOT Large Load Interconnection Process

Dear Chairman and Commissioners:

The attached presentation provides an overview of ERCOT's Large Load interconnection process, including a review of historical transmission planning criteria and the continued evolution of processes to meet the demands associated with the rapid growth of Large Loads seeking to interconnect to the ERCOT Region. The presentation highlights new challenges and opportunities for continued improvement.

ERCOT representatives will be available at Thursday's open meeting to discuss this presentation.

Respectfully,

*/s/ Kristi Hobbs*
Kristi Hobbs
Vice President, System Planning and Weatherization
Electric Reliability Council of Texas, Inc.

# Large Load Interconnection Process Overview

*Kristi Hobbs*
Vice President, System Planning and Weatherization

Public Utility Commission of Texas Open Meeting
October 23, 2025

# Overview

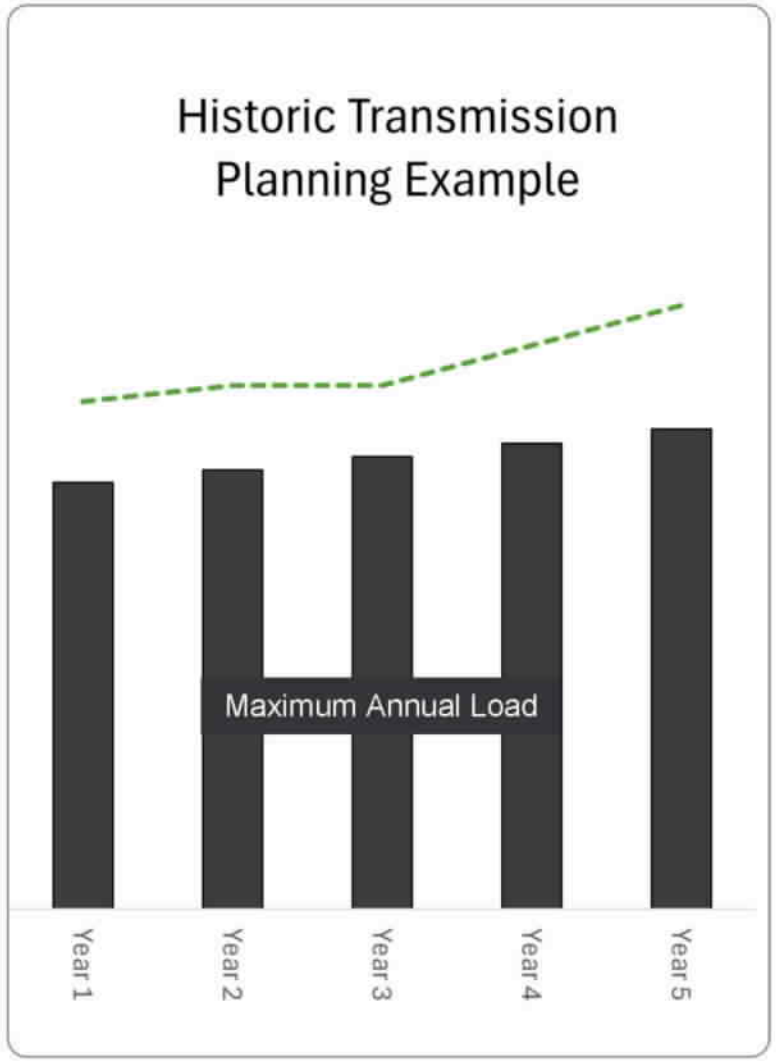
- **Purpose**
  Review historical transmission planning criteria and how it continues to evolve to meet the demands resulting from the rapid explosion of Large Loads, which can be constructed faster than traditional transmission planning timelines. Discuss challenges as well as opportunities for continued improvement to reliably meet the demands of ERCOT grid.

- **Key Takeaways**
  - Historic transmission planning rules allowed for timely transmission build out to meet consistent, predictable demand growth. A minimal transmission capacity margin, which was sufficient to absorb forecast variances, balanced transmission costs to Texas consumers.
  - The volume and changing characteristics of Large Loads seeking ERCOT interconnections have uncovered new challenges which require planning processes to evolve to reliably connect Load.
  - Although the ERCOT Region can build transmission faster than any other region, bridge solutions should be considered in the nearer term until system upgrades can be completed to continue to facilitate the economic development that is powering Texas' future.

# ERCOT's Role in Grid Planning

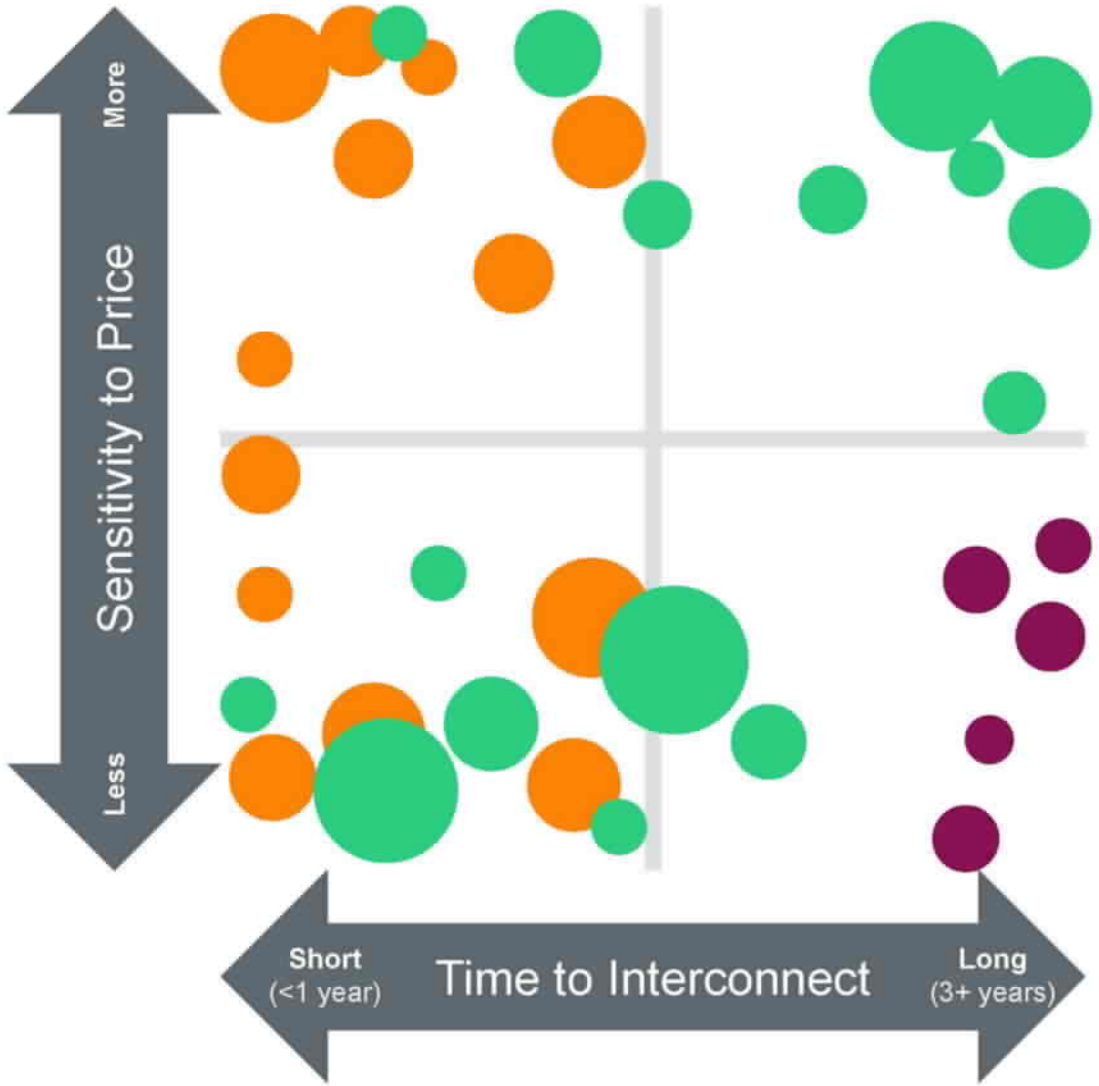
- ERCOT, in coordination with Transmission Service Providers (TSPs), analyzes future grid conditions to ensure both existing and future Load can be reliably served at all times.
- Historic demand growth was fairly predictable. Traditional Large Loads were slower to construct and consume power. Increased demand could be accommodated via longer term transmission planning processes.
- As more Inverter Based Resources are integrated into the system and Large Loads can develop on faster timelines, historical transmission capacity margins can no longer absorb planning forecast variances.
- Transmission limits associated with serving Load vary according to system conditions (i.e., generation and transmission outages, variability of wind and solar output, Energy Storage Resource charging status, level of demand).
- ERCOT's role is to recommend transmission improvements to ensure Load never exceeds the region's ability to supply that Load *at any time*.

**Key Takeaway:** Historic transmission planning rules allowed for timely transmission build to meet consistent demand growth. It allowed for a minimal transmission capacity margin which was sufficient to absorb forecast variances and balanced transmission costs to consumers.




*Green Line represents most limiting scenario where some outages may be in place, wind and solar generation in the region are minimized, and batteries in the region are discharging. Limiting condition may also only exist a few hours each year.*

# Changing Characteristics of Large Loads Coming to ERCOT




## Historical Large Loads

- Typically industrial facilities
- Long timelines to interconnect can be studied by traditional planning processes
- Little price-sensitive behavior in real-time

## Initial Wave of Large Loads

- Mostly crypto-mining, data centers (traditional and AI), some oil field electrification
- Much shorter timeline to interconnect (months rather than years)
- Some Loads are extremely sensitive to price

## Current Future Large Load Interests

- AI data centers, some crypto-mining, hydrogen/electrofuel production
- Range of interconnection timelines and price sensitivity

**Key Takeaway:** Current Large Loads seeking interconnection may have faster development timelines and varying price sensitivities.

# Interim Interconnection Process for Large Loads

**Phase 1**
Interconnection Request and Planning Studies

- Interconnection request made to TSP
- ERCOT and TSP agree to study scope and assumptions
- TSP performs reliability studies and identifies upgrades needed to serve the Load
- ERCOT reviews studies and establishes the amount of Load that can be served reliably
  - To achieve the full requested Load, transmission upgrades may be necessary

**Phase 2**
Agreements and Modeling

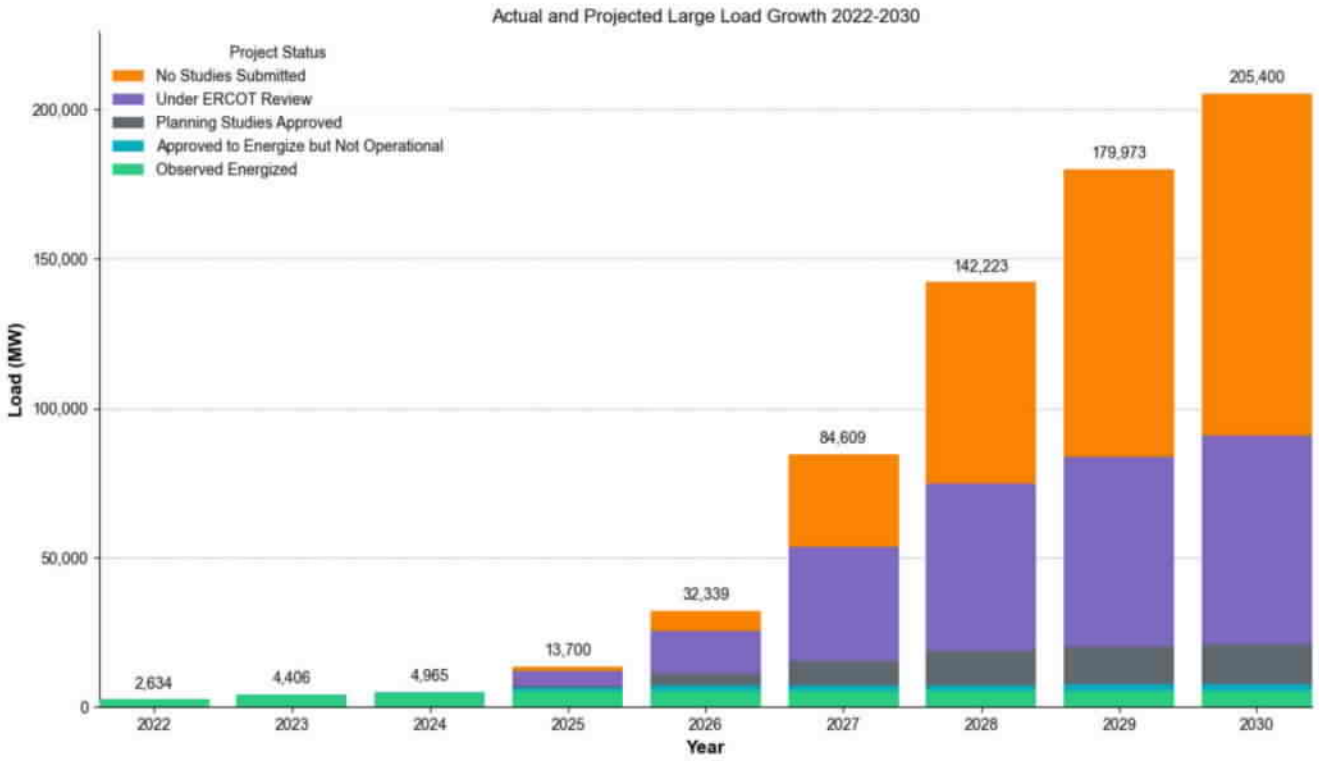
- Customer signs a binding agreement with TSP
- Load is added to ERCOT's planning models and Network Operations Model
- ERCOT Interim Ride-Through Evaluation (*introduced June 2025* - must be completed before Approval to Energize)

**Phase 3**
Approval to Energize

- TSP (or RE for netted loads) requests ERCOT approval to energize the Load
- ERCOT validates Load modeling and telemetry quality
- Load is approved to consume energy up to established ERCOT limit
- After energization, ERCOT monitors Load consumption to ensure limits identified in planning studies are not exceeded

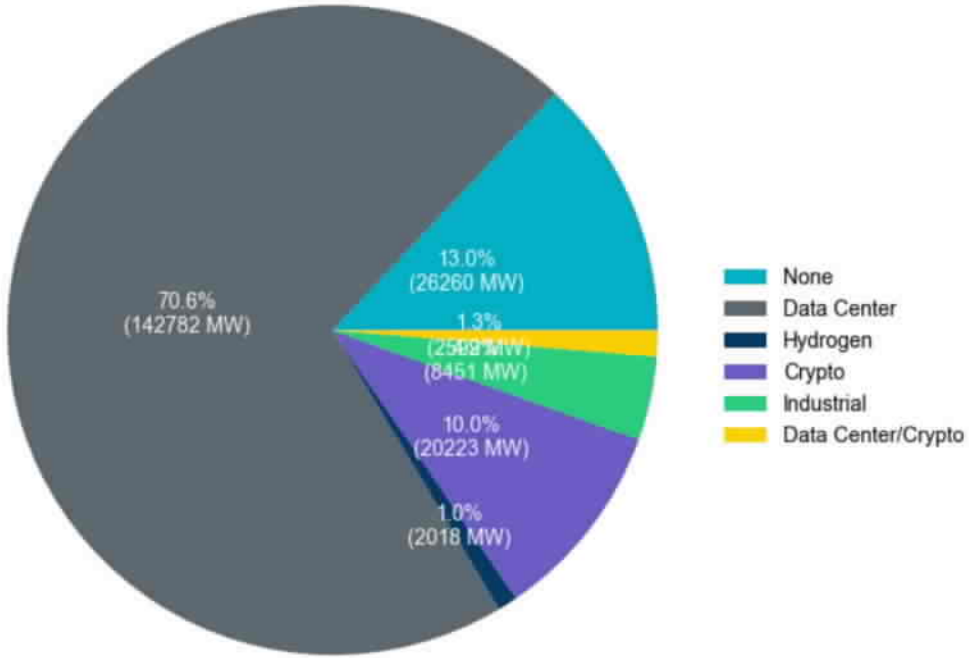
**Key Takeaway:** Interim process was established envisioning approximately 40-50 price sensitive Large Loads that could be studied individually and managed in real-time like generation.

# Large Load Interconnection Requests as of October 14, 2025




| Project Status | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 |
|---|---|---|---|---|---|---|---|---|---|
| No Studies Submitted | 0 | 0 | 0 | 1,714 | 6,995 | 30,886 | 67,232 | 96,116 | 114,246 |
| Under ERCOT Review | 0 | 0 | 0 | 4,720 | 13,875 | 38,461 | 56,379 | 63,767 | 70,162 |
| Planning Studies Approved | 0 | 0 | 0 | 637 | 4,268 | 8,061 | 11,411 | 12,589 | 13,491 |
| Approved to Energize but Not Operational | 0 | 100 | 154 | 1,395 | 1,967 | 1,967 | 1,967 | 2,267 | 2,267 |
| Observed Energized | 2,634 | 4,306 | 4,811 | 5,234 | 5,234 | 5,234 | 5,234 | 5,234 | 5,234 |
| **Total (MW)** | **2,634** | **4,406** | **4,965** | **13,700** | **32,339** | **84,609** | **142,223** | **179,973** | **205,400** |

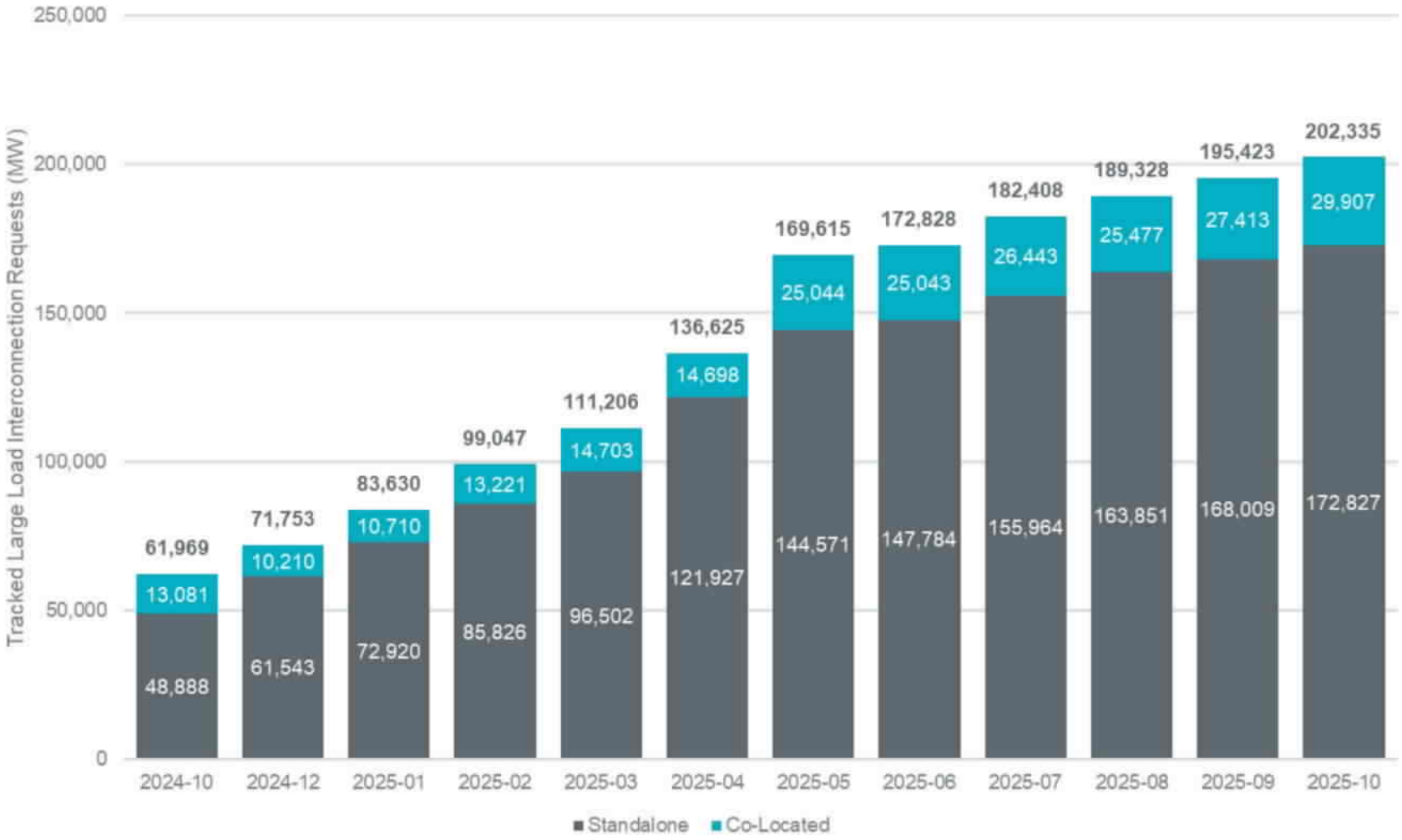
## Large Loads by Project Type




**Key Takeaway:** ERCOT is tracking approximately 205 GW of Large Loads seeking interconnection (compared to 56 GW in September 2024) of which ~70% are data centers.

# Large Load Requests – Past 12 Months

- ERCOT continues to see rapid growth in Large Load interconnection requests
- Many interconnection requests exceed 1 GW *per site*
- Both transmission and resource adequacy should be considered in how quickly Large Loads can connect and ramp up
- PUC load forecasting rules will be instrumental in identifying credible Loads that should be planned for




**Key Takeaway:** The amount of Load seeking interconnection by 2030 has grown by 140 GW (227%) in the past twelve months.

# Challenges Manifesting in Each Phase of the Current Process

**Phase 1**
Interconnection Request and Planning Studies

**Challenges**
- Many customers are seeking connection in specific areas (WTX, DFW, etc.). Study assumptions may quickly become outdated as additional customers arrive in an area.
- There are many unsigned "officer letter" loads in the planning cases. This makes studies difficult because it is not known which of these loads will connect (issues identified in studies may or may not be real).
- Mismatch between assumptions in LLIS and other ERCOT planning studies – can be difficult to get transmission upgrades approved.

**Phase 2**
Agreements and Modeling

**Challenges**
- TSPs can sign agreements with customers before studies are completed. This creates an expectation from the customer that the signed Load can be served.

**Phase 3**
Approval to Energize

**Challenges**
- Requirements are evolving as ERCOT better understands the risks posed by these Large Loads. This can lead to new requirements that are identified after customer has already started construction.
- Communication gap - approval of planning studies is not an approval to energize

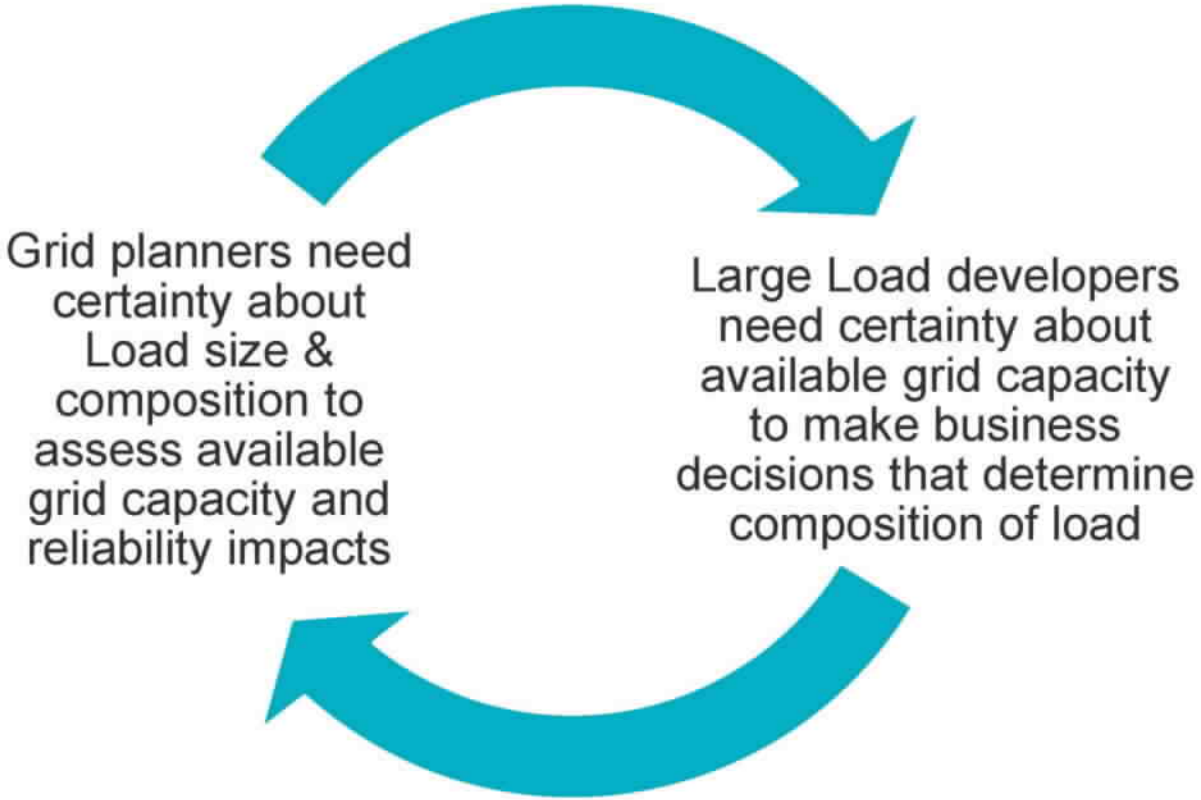
#1 Studies often require updates if other area Loads sign agreements

#2 Studies may need to be updated as the industry recognizes new reliability risks associated with these types of Loads

**Key Takeaway:** With an open access transmission system where transmission capacity is not reserved, restudies can be required as new Loads seek to be interconnected.

# Challenge #1 – Individual Study Approach and Restudies

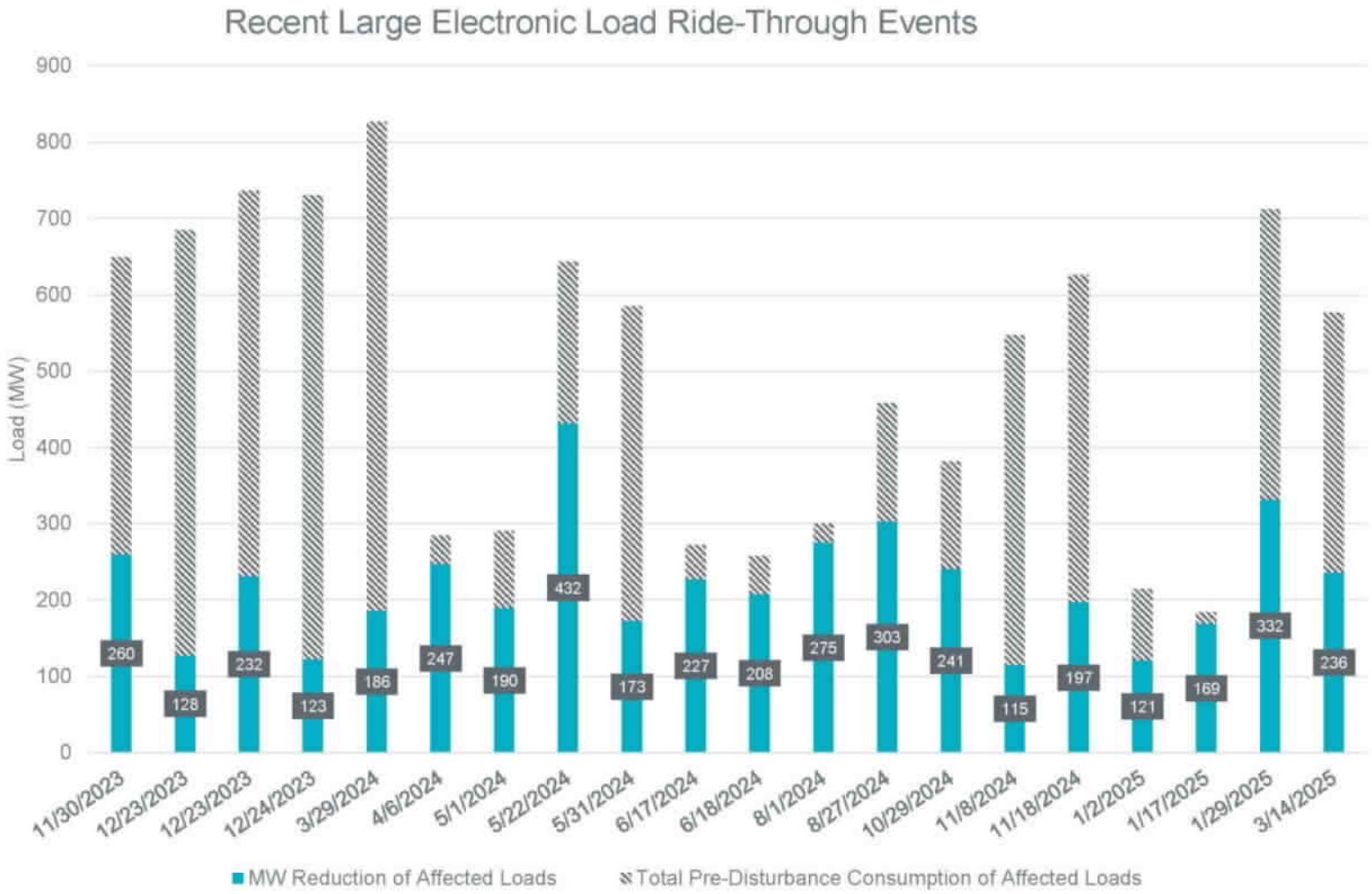
- Both planners and Load need certainty during the planning phase
- Lack of information requires assumptions
  - Which Loads to include in study area
  - Load composition and dynamic response
- Small changes in study assumptions and Load model parameters can dramatically change study outcomes
- Uncertainty and volume of requests is straining the individual study approach




**Key Takeaway:** Revised approaches which consider multiple Large Loads in a single study could help alleviate restudies.

# Challenge #2 – Ride-Through Behavior (New Reliability Risk)

- ERCOT continues to see Load reduction for even shallow voltage dip events
- This behavior represents a growing risk to the stability of grid frequency as more Large Loads connect
- Current models do not capture this behavior, making accurate studies to assess risk and identify transmission improvements difficult



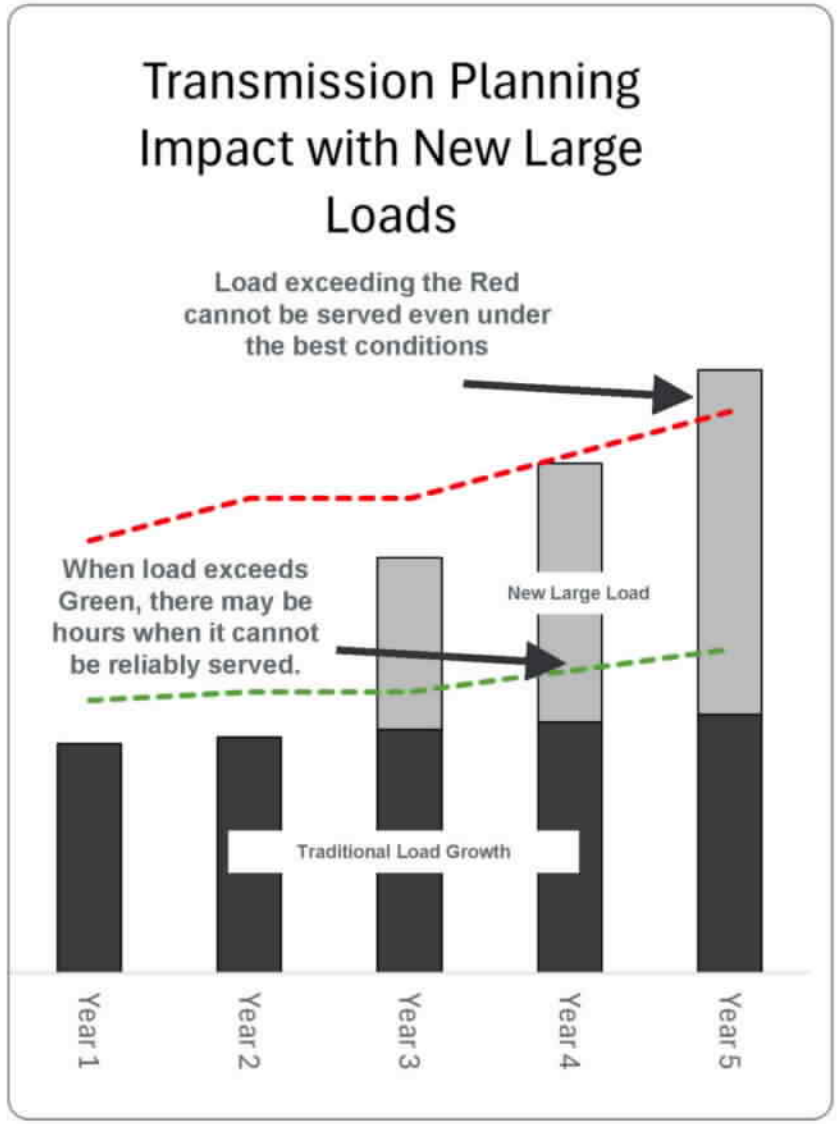

**Key Takeaway:** Reliability risks must be studied and planned for to reliably operate the grid.

# Opportunities for Rapidly Evolving Large Loads

**ERCOT incorporates Large Load requests based on input from the TSPs connecting the new Large Loads in addition to the existing Load growth.**

- Current transmission planning rules require ERCOT to plan for firm Load and ensure the ability to serve Load under any conditions *at any time*.
- With an evolving grid, Large Loads that are willing to respond to system conditions could have an opportunity to interconnect if they can adjust their consumption until transmission and/or resource additions can be incorporated on the system.

**Key Takeaway:** Large Loads which are flexible could utilize available transmission capacity if they are willing to curtail under certain conditions.

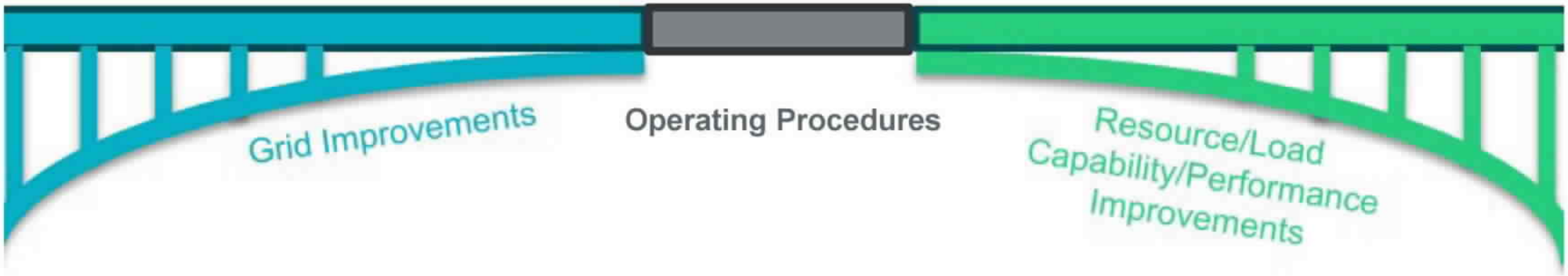



Red Line represents best-case scenario for region with no outages, wind and solar generation are maximized, and batteries are not charging. This optimal condition may only exist a few hours each year.

# Large Load Interconnection Process – Possible Options

- Consider system-wide or regional studies to eliminate restudies
- Prioritize Large Loads that can positively impact both transmission and resource adequacy
  - Registration as a Controllable Load Resource (CLR) – dependent upon implementation of NPRR1188/NPRR1244
  - Credit for bringing own generation
- Create single source submittal of Large Load information to ERCOT
- Standardization across TSPs via PUC's proposed load forecast and load interconnection standards
- Re-evaluate whether direct relationship is needed between ERCOT & Large Load customers to increase transparency and efficiency in process

**Key Takeaway:** Bridge solutions will be needed to support Large Load growth in the near term. ERCOT will work with the PUC and the Large Load Working Group to continue evolving the process.

# Solutions for Rapidly Evolving Large Loads

**Grid, Resource and Load improvements bridged with updated ERCOT operating procedures will position the grid to reliably incorporate Large Load additions.**